

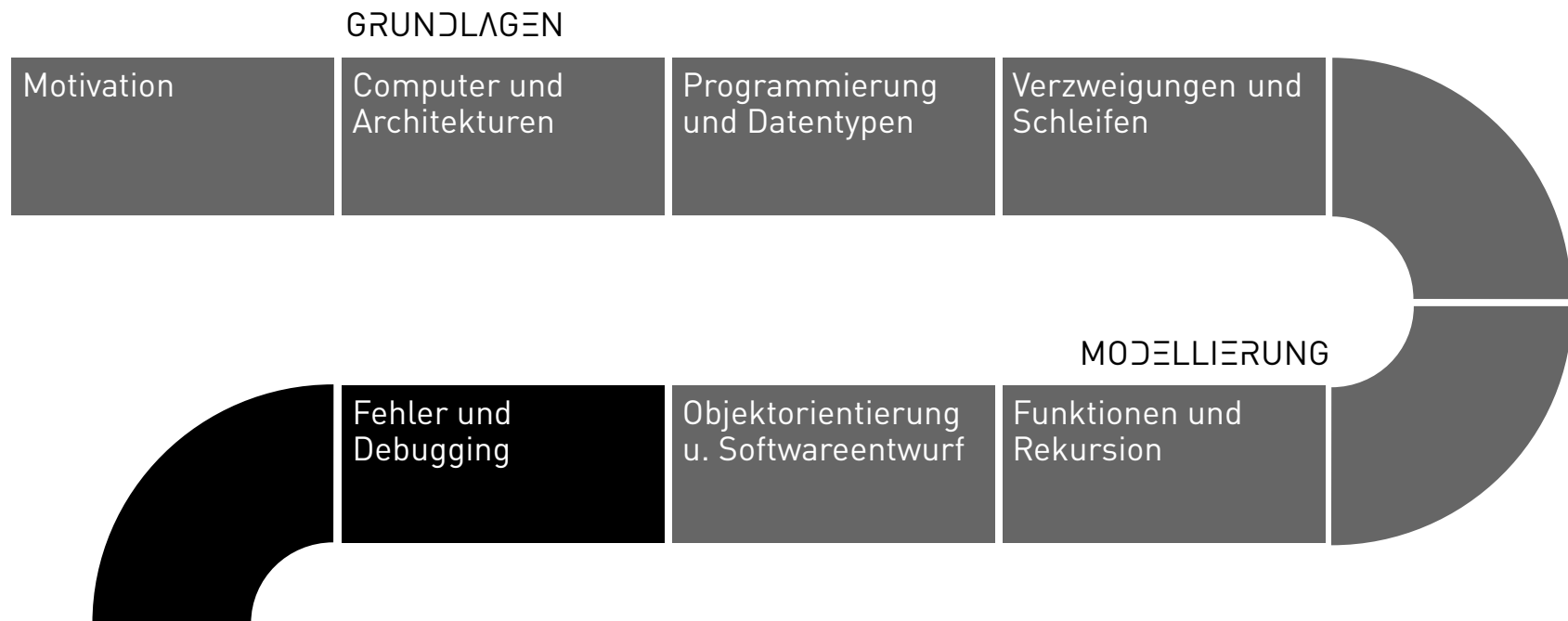
Programming Databases

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Program Flow

MIDJOURNEY: DOVES TO BLOCKS, REF. M. C. ESCIER

PROCESS



PROGRAM FLOW - EXAMPLE

```
In [ ]: x = 3          # Assignment
        y = 4          # Assignment

        def pythagoras(a, b):    # Storing the function
            return math.sqrt((a*2)+(b*2))  # Calculation

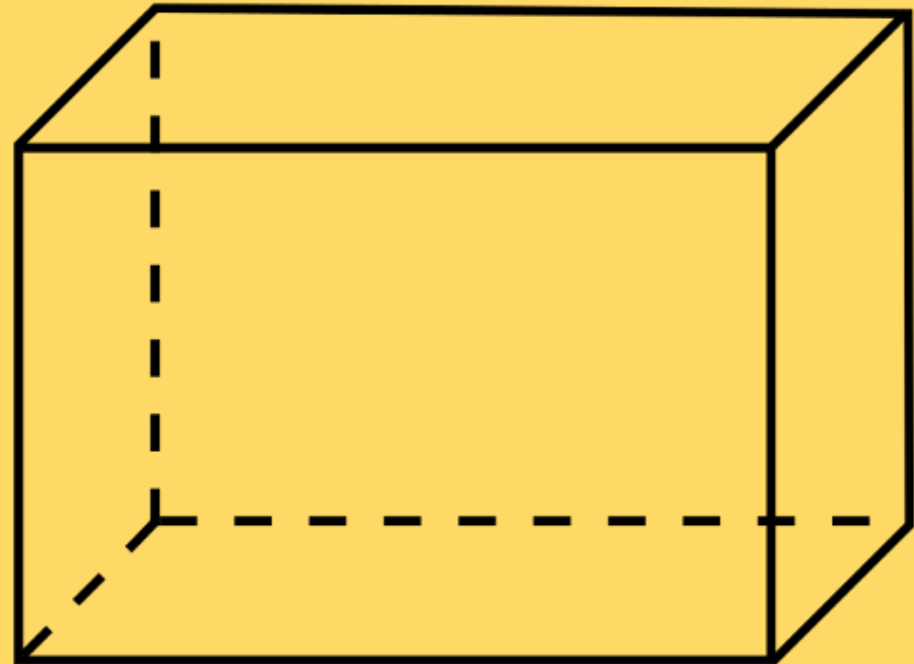
        z = pythagoras(x, y)    # Function call, assign to z
        print(z)                # Variable output
```

DEFINING THE PROBLEM

- The most important skill in programming is to translate a real-world problem into individual program instructions (i.e., operations on symbols).
- *First step:* What are the inputs, what are the outputs of the current problem?
- Not unlike how a function is defined!
 - Input parameters
 - Return value

LECTURE HALL QUESTION

What inputs and outputs are involved in calculating the volume of a rectangular prism (house)?



EXAMPLE: VOLUME CALCULATION OF A RECTANGULAR PRISM (HOUSE)

- *Simplification:* The house is rectangular and has a flat roof.
- *Inputs:* Side lengths **a** and **b**, height **h** (in m)
- *Outputs:* Volume in m^3
- *First considerations:*
 - What data types do these inputs and outputs have?
 - What different scenarios exist?

BREAKING THE PROBLEM INTO STEPS

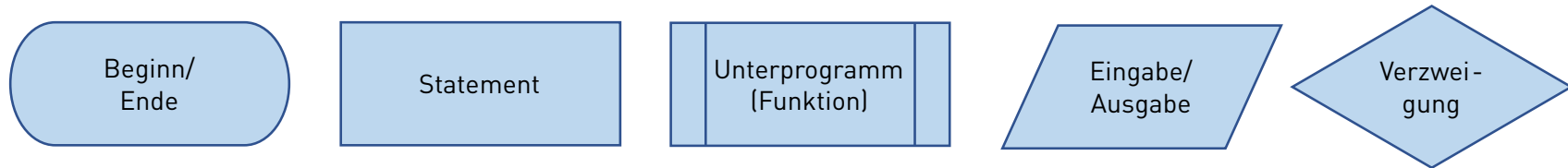
- *The core question:*

"How do we get from the given input to the desired output?"

- *Not Easy:* To answer this question for arbitrary problems, one must adopt a way of thinking, break the problem into steps, and work through them.
- *Our Advantage:* The ability to decompose and solve problems is a fundamental skill for engineers.

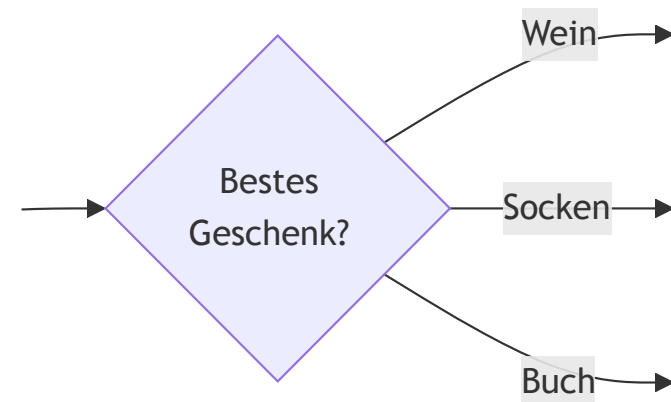
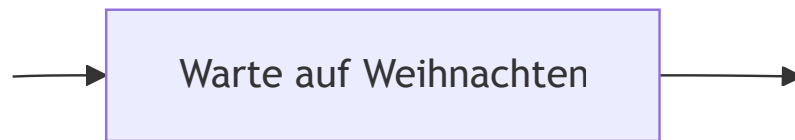
A TOOL FOR DESIGNING ALGORITHMS

- Also called '*flowcharts*'
- Describe sequences of operations
- Can have the following elements, which are connected by arrows:

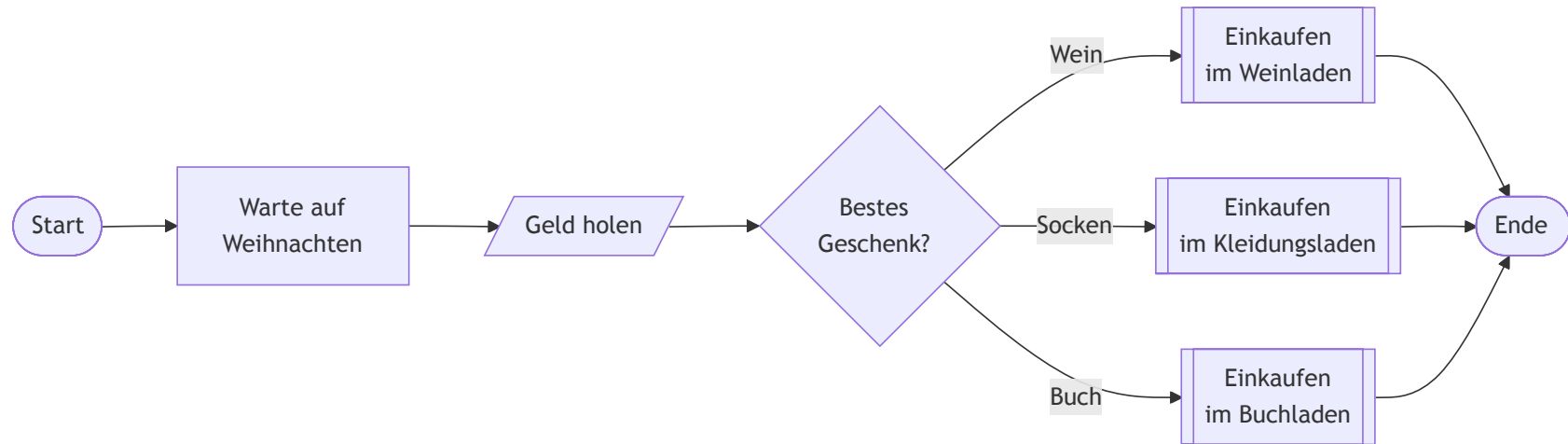


PROGRAM FLOWCHARTS — FUNDAMENTALS

- The most important blocks:
- Statements do not need to be written in the syntax of a programming language – they can be pseudocode.
- Arrows originate from a branching point, indicating whether the condition evaluated to True or False (if/else).



EXAMPLE: CHRISTMAS SHOPPING



WHAT IS THIS USED FOR IN EVERYDAY LIFE?

©Midjourney: Dorian Gray

Process modeling

- Geothermal process flows
- IP management process optimization
- Business processes
- Quality assurance

Patents

- Algorithmic descriptions
- Process patents
- Technical documentation

LESSONS LEARNED - FLASHCARDS

- The question goes on the front of the flashcard, the answer on the back
- The cards for Subject 1 are reviewed **daily**
- The Subject 2 cards are reviewed **every other day**
- The Subject 3 cards are reviewed **once a week**
- and the Subject 4 cards are reviewed **every 30 days**

Questions?

